

Sequences & Series



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 6$ and common difference $d = 4$, find a_{12} .
 - 2 For the arithmetic sequence with $a_1 = 20$ and $d = -3$, find a_{15} .
 - 3 A geometric sequence has $a_1 = 3$ and each term is 2 times the previous one. Find the smallest n for which $a_n = 384$.
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Arithmetic series (sum formula)

- 4 Find S_{12} for the arithmetic sequence with $a_1 = 6$, $d = 4$ (using $a_{12} = 50$ from above).
 - 5 Find S_8 for the arithmetic sequence with $a_1 = 2$, $d = 5$.
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Geometric sequences

- 6 For the geometric sequence with $a_1 = 4$ and common ratio $r = 3$, find a_6 .
 - 7 A geometric sequence begins 5, -10 , 20, -40 , Find r and a_7 .
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Geometric series (finite)

- 8 Find S_6 for the geometric sequence with $a_1 = 4$, $r = 3$ (using $a_6 = 972$ from above).
 - 9 Find S_5 for the geometric sequence 5, -10 , 20, -40 ,
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Infinite geometric series

- 10 Find the sum of the infinite geometric series $20 + \frac{20}{3} + \frac{20}{9} + \dots$.
- 11 Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \dots$.
- 12 Explain why the infinite geometric series with $a = 5$, $r = 2$ has no finite sum.